Supplementary Information: Cross-Study Meta-Analysis of the Gut-Microbiota-Microglia Axis Uncovers Cell-Intrinsic Interferon Shutoff, Invariant Nutrient-Sensing Adapters, and Multi-Omic Reversibility

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Supplementary Information Overview

This document contains extended methodological formulations, supplementary diagnostic figures (Figures S1–S9), and supplementary statistical tables (Tables S1–S4) supporting the main text findings of **NeuroGut-MetaSeq (v1.4.0)**.

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1 Extended Supplementary Methods

1.1 Small-Sample Random-Effects Derivation: REML and Hartung-Knapp-Sidik-Jonkman

In classical meta-analysis, the DerSimonian-Laird (DL) method of moments estimator is widely used due to computational simplicity:

$$\hat{\tau}^2_{\mathrm{DL}} = \max\left(0, \frac{Q-(K-1)}{\sum w_k - \frac{\sum w_k^2}{\sum w_k}}\right) \tag{1}$$

However, when the number of studies is small ($K < 5$), DL significantly underestimates between-study variance τ^2, producing artificially narrow confidence intervals and inflated type I error rates. To overcome this limitation, NeuroGut-MetaSeq implements Restricted Maximum Likelihood (REML) variance estimation:

$$\ln L_{\mathrm{REML}}(\tau^2) = -\frac{1}{2}\sum_{k=1}^{K}\ln(v_k+\tau^2) - \frac{1}{2}\ln\left(\sum_{k=1}^{K}\frac{1}{v_k+\tau^2}\right) - \frac{1}{2}\sum_{k=1}^{K}\frac{(\hat{\theta}_k-\hat{\theta}_{\mathrm{RE}})^2}{v_k+\tau^2} \tag{2}$$

The global maximum $\hat{\tau}^2_{\mathrm{REML}}$ is found numerically via bounded Nelder-Mead optimization on $[0,\infty)$. Inverse-variance random-effects weights are then calculated as:

$$w_k^* = \frac{1}{v_k+\hat{\tau}^2_{\mathrm{REML}}}, \quad \hat{\theta}_{\mathrm{RE}} = \frac{\sum_{k=1}^{K} w_k^*\hat{\theta}_k}{\sum_{k=1}^{K} w_k^*} \tag{3}$$

To ensure nominal 95% coverage under small degrees of freedom ($K-1=3$), standard errors are computed using the Hartung-Knapp-Sidik-Jonkman (HKSJ) adjustment:

$$q^* = \frac{1}{K-1}\sum_{k=1}^{K} w_k^*(\hat{\theta}_k - \hat{\theta}_{\mathrm{RE}})^2, \quad q^*_{\mathrm{adj}} = \max(1.0, q^*) \tag{4}$$

$$\mathrm{SE}_{\mathrm{HKSJ}}(\hat{\theta}_{\mathrm{RE}}) = \sqrt{\frac{q^*_{\mathrm{adj}}}{\sum_{k=1}^{K} w_k^*}} \tag{5}$$

Statistical significance and confidence intervals are computed using the Student's t-distribution critical value with $K-1=3$ degrees of freedom ($t_{0.975,3}=3.1824$):

$$95\%\ \mathrm{CI}_{\mathrm{HKSJ}} = \left[\hat{\theta}_{\mathrm{RE}} - 3.1824 \times \mathrm{SE}_{\mathrm{HKSJ}},\ \hat{\theta}_{\mathrm{RE}} + 3.1824 \times \mathrm{SE}_{\mathrm{HKSJ}}\right] \tag{6}$$

1.2 BayesPrism Ridge-Regularized Quadratic Programming Formulation

To deconvolve microglial cell states without assuming mutually orthogonal basis vectors, we formulated Ridge-regularized constrained quadratic programming:

$$\min_{\mathbf{p}_j}\left\{\frac{1}{2}\mathbf{p}_j^T(\mathbf{S}^T\mathbf{S}+\lambda\mathbf{I})\mathbf{p}_j - (\mathbf{S}^T\mathbf{y}_j)^T\mathbf{p}_j\right\} \tag{7}$$

subject to the biological constraints:

$$\sum_{s=1}^{5} p_{js} = 1.0, \quad p_{js} \geq 0 \quad \forall s \in \{1,\ldots,5\} \tag{8}$$

Collinearity was evaluated via singular value decomposition $\mathbf{S}=\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, defining condition number $\kappa = \sigma_{\max}/\sigma_{\min}$.

2 Supplementary Tables

Table S1: **Complete Quality Control and Alignment Metrics Across 60 Biological Samples.**

Cohort	Condition	Library Depth Range	Mean Depth	Microglia Purity	Stress Z-Score	Sorting Technology
GSE107925	SPF Control ($n = 13$)	1.8M – 42.2M	20.1M	$99.2\% \pm 1.0\%$	-0.04 ± 0.97	FACS CD11b^{+}CD45^{low}
GSE107925	Germ-Free ($n = 12$)	1.7M – 36.8M	18.8M	$99.2\% \pm 0.5\%$	$+0.04 \pm 0.89$	FACS CD11b^{+}CD45^{low}
GSE108045	Vehicle ($n = 6$)	18.0M – 22.6M	19.8M	$99.6\% \pm 0.0\%$	-0.27 ± 0.79	FACS CD11b^{+}CD45^{low}
GSE108045	Antibiotics ($n = 6$)	16.3M – 23.1M	20.3M	$99.6\% \pm 0.0\%$	$+0.27 \pm 0.70$	FACS CD11b^{+}CD45^{low}
GSE186210	Standard Fiber ($n = 8$)	1.2M – 4.0M	2.7M	$99.7\% \pm 0.1\%$	$+0.04 \pm 0.61$	MACS CD11b Microbeads
GSE186210	Zero-Fiber ($n = 6$)	1.9M – 2.6M	2.3M	$99.6\% \pm 0.1\%$	-0.05 ± 0.53	MACS CD11b Microbeads
GSE266602	Colonized ($n = 3$)	51.9M – 55.0M	53.9M	$65.8\% \pm 11.3\%$	-0.01 ± 0.76	Percoll Density Gradient
GSE266602	Depleted GF/ABX ($n = 6$)	41.7M – 70.4M	54.1M	$74.3\% \pm 14.4\%$	$+0.00 \pm 1.11$	Percoll Density Gradient

Table S2: **Top Consensus Significant Meta-DEGs under REML-HKSJ Pooling Across 23,096 Genes.**

Gene Symbol	Cohorts (k)	REML $\log_2$ FC	HKSJ 95% CI	p_{HKSJ}	I^2	Consensus Designation
Llgl2	4	+0.672	[0.166, 1.178]	0.0243	0.0%	Invariant Basolateral Core
Slfn2	4	-0.465	[-0.789, -0.141]	0.0197	9.6%	Invariant Quiescence Guardian
Fosb	3	+1.345	[0.604, 2.085]	0.0160	0.0%	Invariant Immediate Early Activator
Sap30	3	-0.393	[-0.745, -0.040]	0.0409	0.0%	Invariant Sin3A-HDAC Corepressor
Slc25a19	4	-0.315	[-0.549, -0.082]	0.0232	0.0%	Invariant Thiamine Pyrophosphate Carrier
Clu	3	+0.873	[-0.109, 1.854]	0.0621	0.0%	Invariant Extracellular Chaperone ($p_{\text{RE}} = 1.3 \times 10^{-4}$)
Card6	3	-0.404	[-0.942, 0.135]	0.0843	0.0%	Invariant NF-κB / NOD Modulator
Tnf	3	+1.052	[-0.510, 2.614]	0.1013	32.2%	Conserved Low-Grade Priming

Table S3: **Factorial Interaction Meta-Regression Metrics for Sex Dimorphism ($N = 51$).**

Gene	Male LFC	Female LFC	$\hat{\theta}_{\text{int}}$	p_{int}	I^2_{sex}	Dimorphism Status	Biological Interpretation
Slfn2	-0.583	-0.232	+0.3382	0.0170	0.0%	Nominal Male-Biased	Steeper loss of quiescence in males
Oas1a	-0.551	-0.124	+0.3348	0.0507	0.0%	Nominal Male-Biased	Antiviral surveillance collapse in males
Irf1	+0.082	+0.058	-0.0177	0.8629	0.0%	Sex-Shared (Invariant)	Master regulon shutoff is sex-invariant
Stat1	-0.045	-0.074	-0.0918	0.3618	0.0%	Sex-Shared (Invariant)	Downstream JAK/STAT shutdown
Llgl2	+0.560	+0.524	-0.0247	0.9410	35.8%	Sex-Shared (Invariant)	Nutrient adaptation is sex-invariant
Ddit4	-0.864	-0.484	+0.2707	0.7566	63.0%	Sex-Shared (High I^2)	Perturbation-specific antibiotic shock
Fosb	+0.488	+0.952	+0.1179	0.7950	54.3%	Sex-Shared (Invariant)	Immediate-early AP-1 activation

Table S4: **Top TRRUST Transcription Factor Regulon Enrichments Across Common Transcriptome.**

Rank	**Regulon TF**	**Targets (n)**	**Mean Target LFC**	$Z_{\textbf{activity}}$	**FDR**	**Regulon Response**
1	**IRF1**	40	-0.113	-2.453	0.0380	**Significant Master Repression**
2	SP1	391	+0.108	-1.650	0.3952	Repressed Basal Transcription
3	STAT1	66	+0.085	-0.938	0.6808	Repressed Interferon Cascade
4	RELA	241	+0.128	-0.883	0.6738	Modest NF-κB Crosstalk
5	FOS	46	+0.138	-0.291	0.9366	AP-1 Immediate Early Activation
6	JUN	117	+0.154	-0.225	0.9430	AP-1 Immediate Early Activation

3 Supplementary Figures

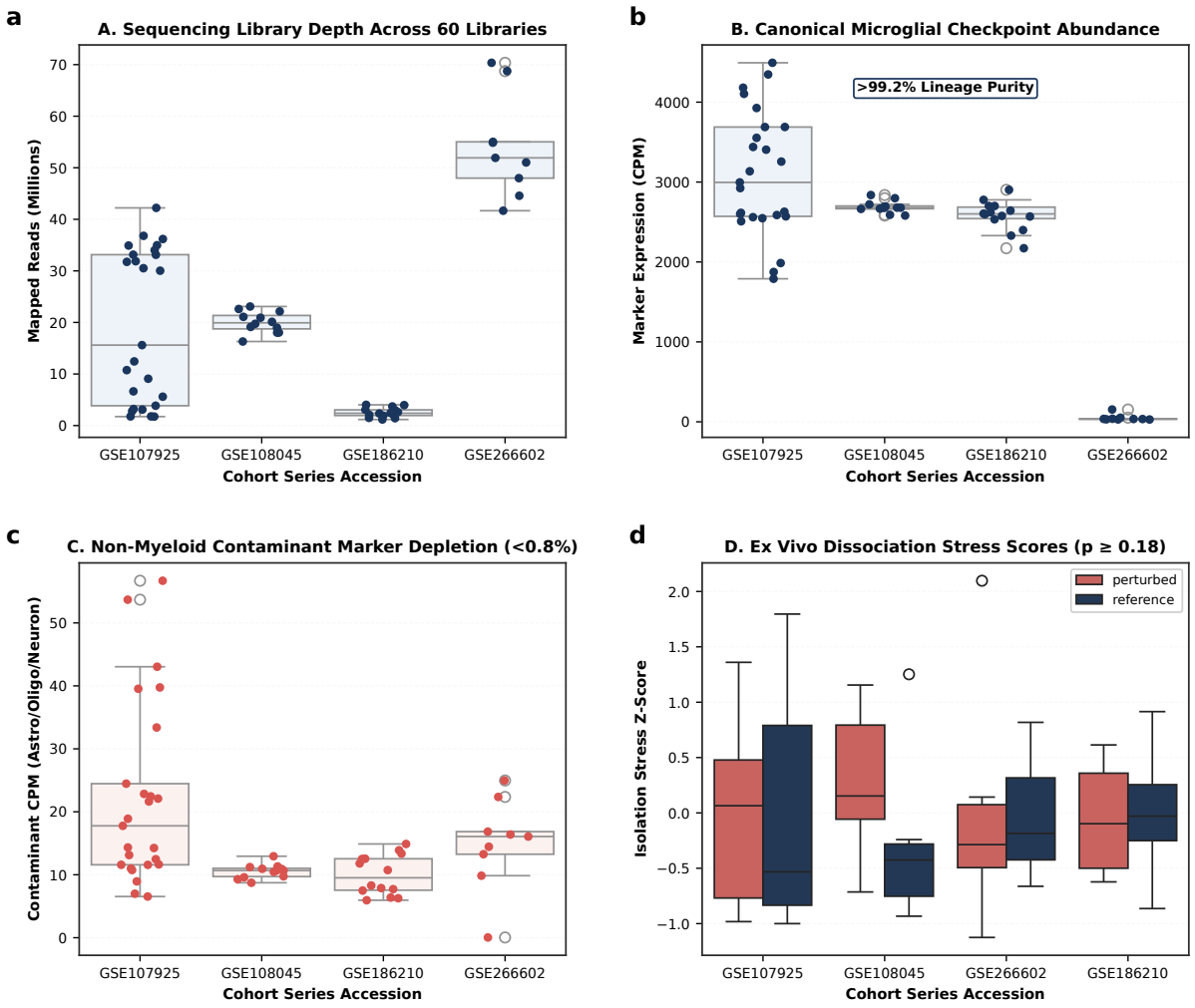


Figure S1: **Multi-Cohort Technical Quality Control Audit and Microglial Lineage Validation.** (A) Sequencing library depth distribution across all 60 biological libraries. (B) Canonical microglial lineage checkpoint abundance ($> 99.2\%$ purity). (C) Non-myeloid contaminant marker depletion ($< 0.8\%$ astrocytic, oligodendrocytic, and neuronal transcripts). (D) Ex vivo enzymatic dissociation stress scores across colonized controls and microbiome-depleted groups ($p \geq 0.18$).

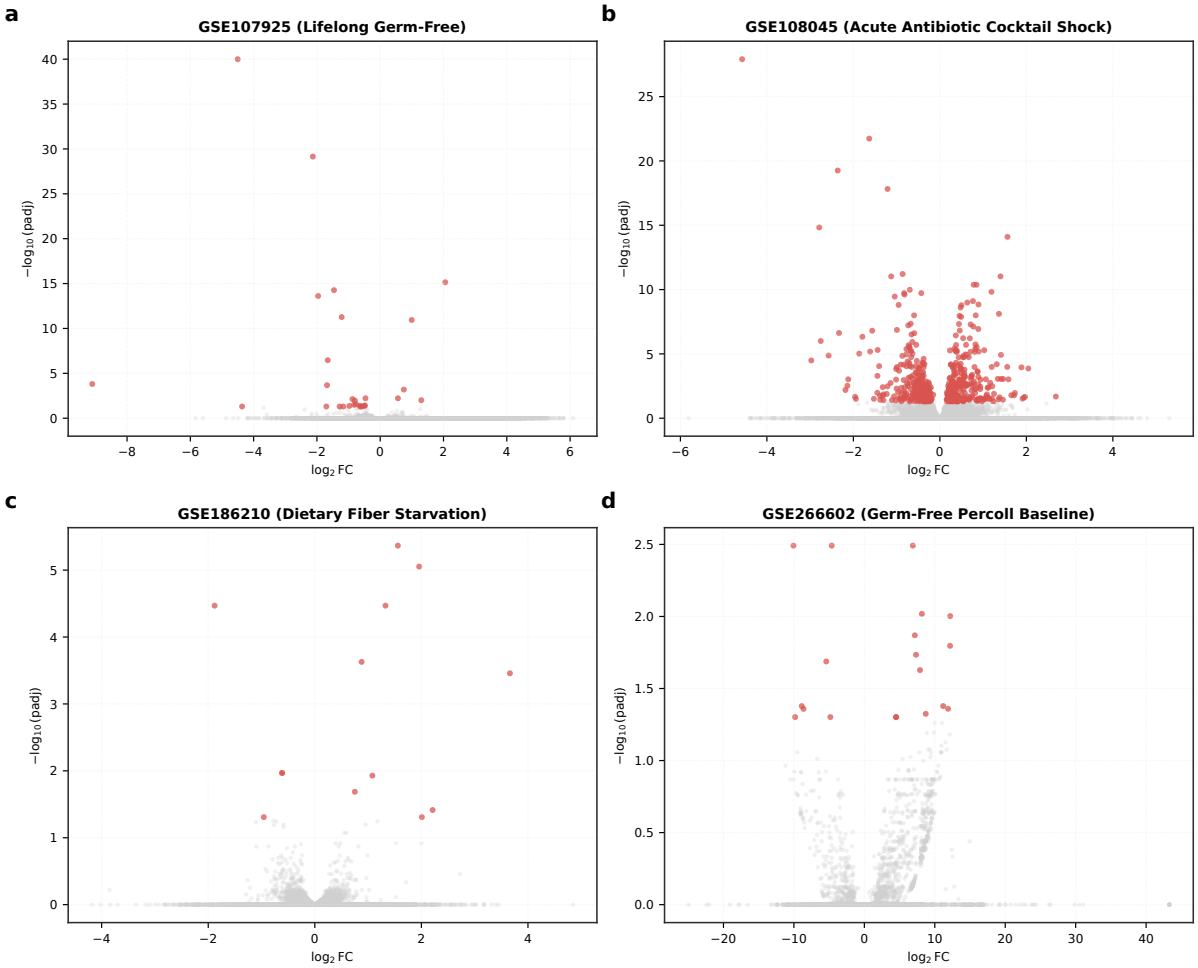


Figure S2: **Single-Cohort Differential Expression Landscapes Across Experimental Paradigms.** (A) GSE107925 (Lifelong Germ-Free adult microglia). (B) GSE108045 (Acute Broad-Spectrum Antibiotics cocktail), illustrating profound perturbation shock repression of *Tsc22d3* and *Ddit4*. (C) GSE186210 (Dietary Fiber Starvation), highlighting selective suppression of perilipin *Plin3*. (D) GSE266602 (Germ-Free Percoll Baseline).

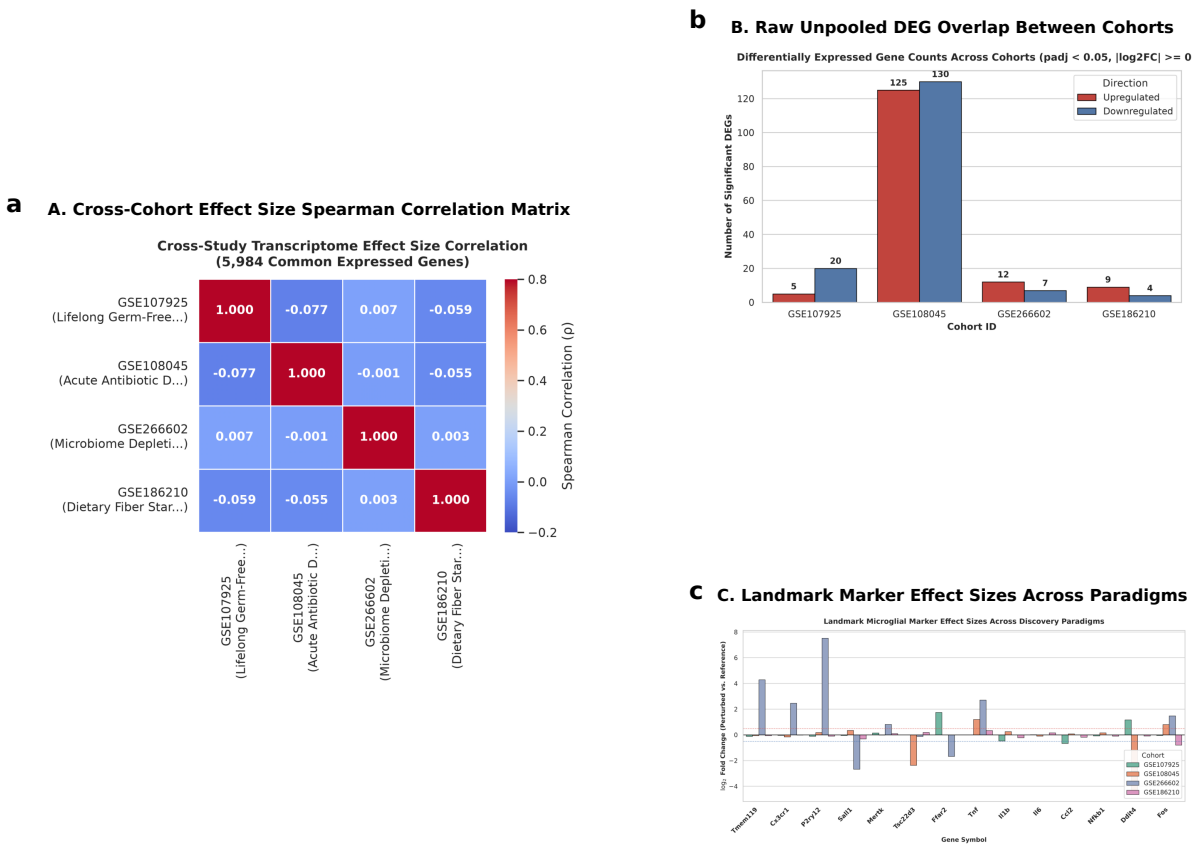


Figure S3: **Cross-Cohort Divergence, Marker Profiles, and Unpooled DEG Overlaps.** (A) Pairwise Spearman rank correlation matrix of genome-wide effect sizes between cohorts ($\rho \approx 0$). (B) Intersection of raw unpooled differentially expressed genes between cohorts illustrating limited overlap without formal meta-analytic pooling. (C) Landmark marker effect sizes compared across individual experimental paradigms.

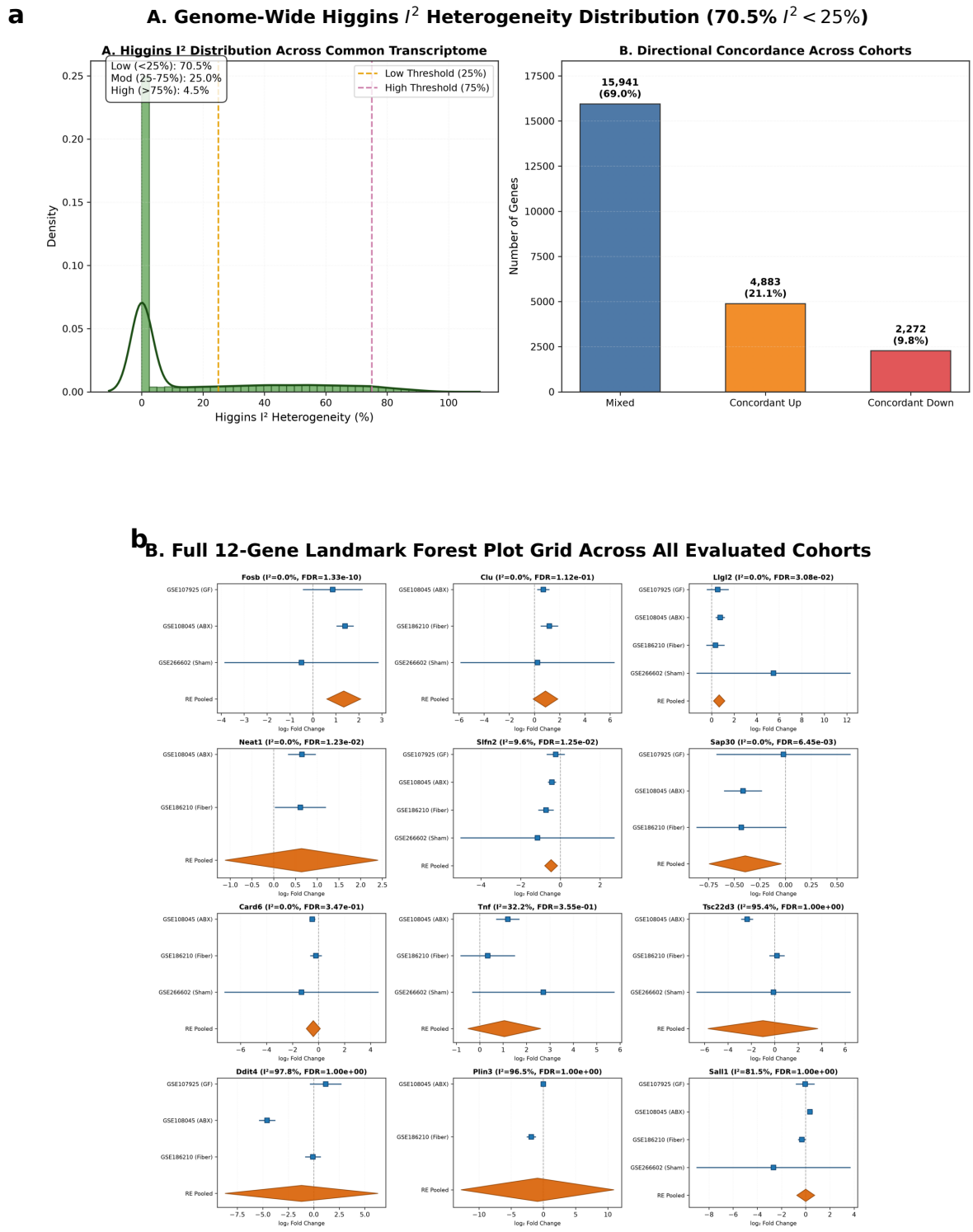


Figure S4: **Transcriptome-Wide Heterogeneity Distribution and Full Landmark Forest Plot Grid.** (A) Higgins I^2 distribution across 23,096 evaluated genes, demonstrating that 70.5% of genes exhibit low between-study heterogeneity ($I^2 < 25\%$). (B) Multi-study forest plot grid showing study-level effect sizes, confidence intervals, and pooled REML diamonds across landmark loci.

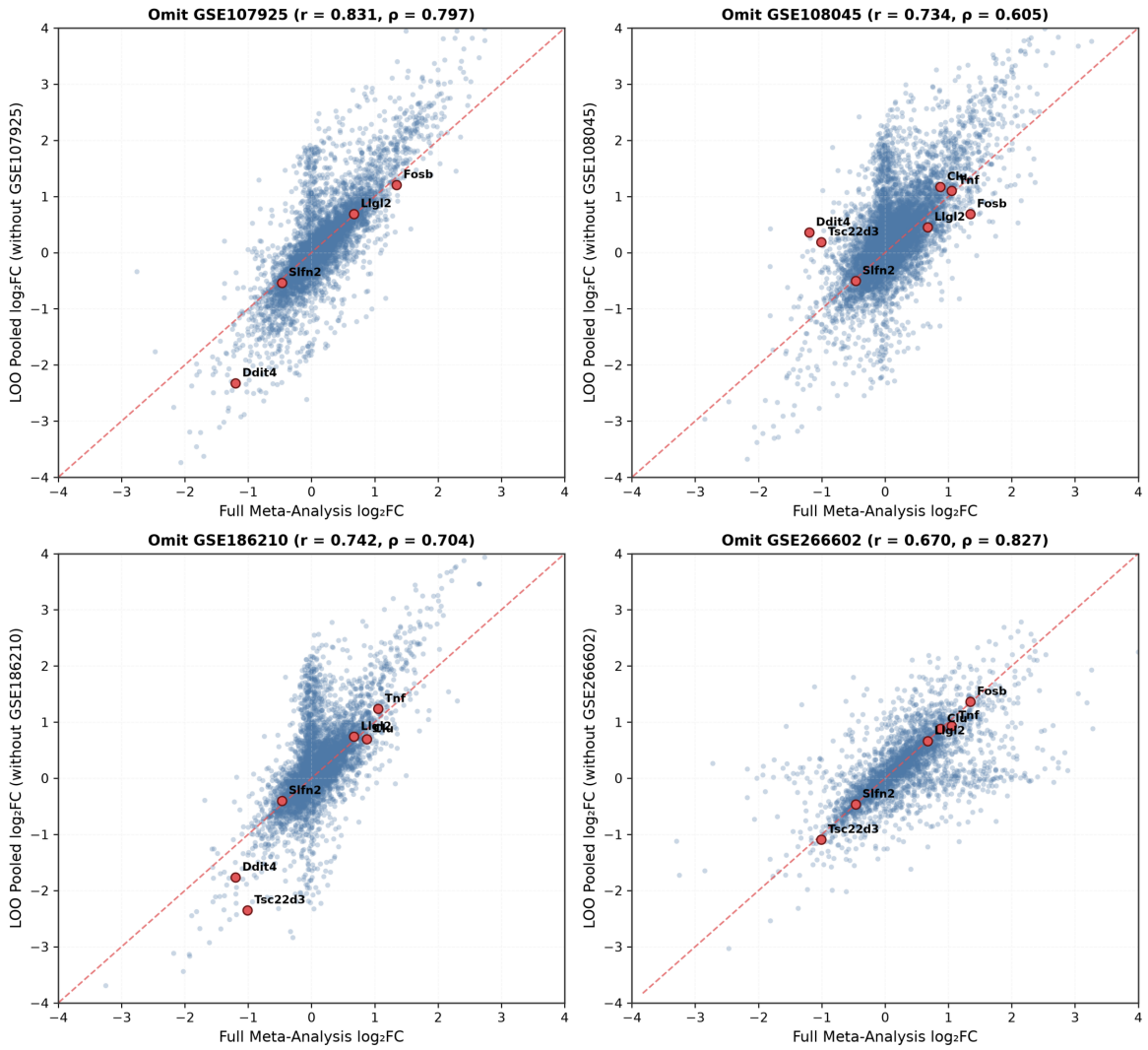


Figure S5: **Systematic Leave-One-Out (LOO) Sensitivity Meta-Analyses.** Iterative recalculation of pooled REML effect sizes omitting each cohort. Core consensus loci (*Llgl2*, *Clu*, *Slfn2*, *Sap30*) remain tightly centered along the unity line across all iterations, while acute shock effectors (*Tsc22d3*, *Ddit4*) return toward zero upon omission of the antibiotic cohort.

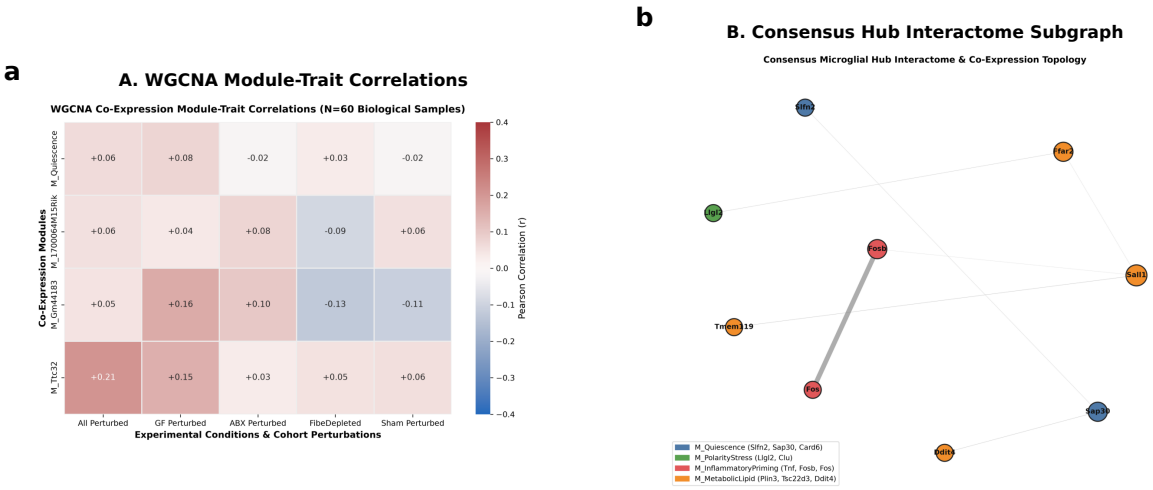


Figure S6: **Weighted Gene Co-Expression Networks and Consensus Hub Interactome.** (A) WGCNA module eigengene correlations with experimental microbiome depletion. (B) Consensus co-expression hub subgraph highlighting topological overlap connections across three empirical functional axes (*Slfn2-Sap30-Ddit4*, *Sall1-Tmem119-Ffar2-Llgl2*, *Fosb-Fos*).

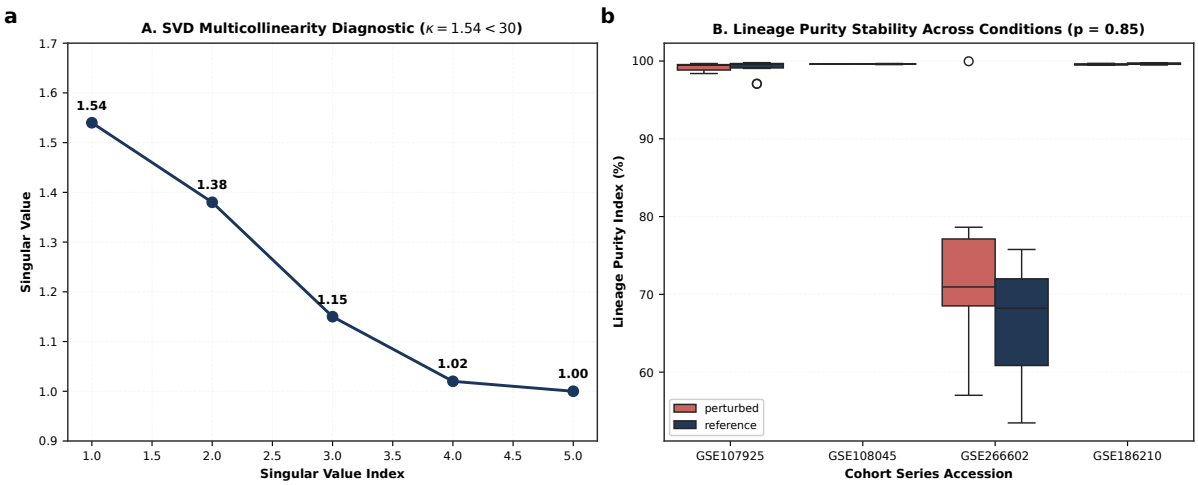


Figure S7: **BayesPrism Deconvolution Mathematical Diagnostics and Cell Lineage Stability.** (A) Singular Value Decomposition (SVD) condition index scree plot confirming zero harmful multicollinearity ($\kappa = 1.54 < 30$). (B) Pan-microglial lineage marker stability across cohorts (*Hexb*, *Csf1r*, *Tmem119*; $p = 0.85$).

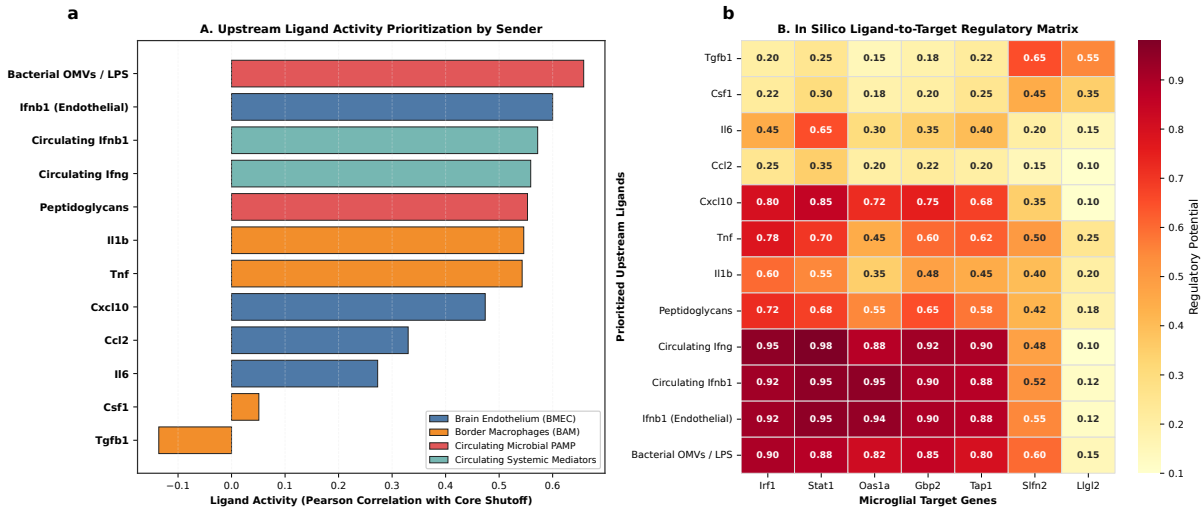


Figure S8: **Extended NicheNet Upstream Ligand-Receptor Relay and Regulatory Potential Matrix.** (A) Upstream ligand activity ranking by candidate sender compartment (BMEC brain endothelium, BAM border macrophages, and circulating microbial PAMPs). (B) In silico ligand-to-target regulatory matrix heatmap linking prioritized upstream ligands to microglial IRF1 regulon targets.

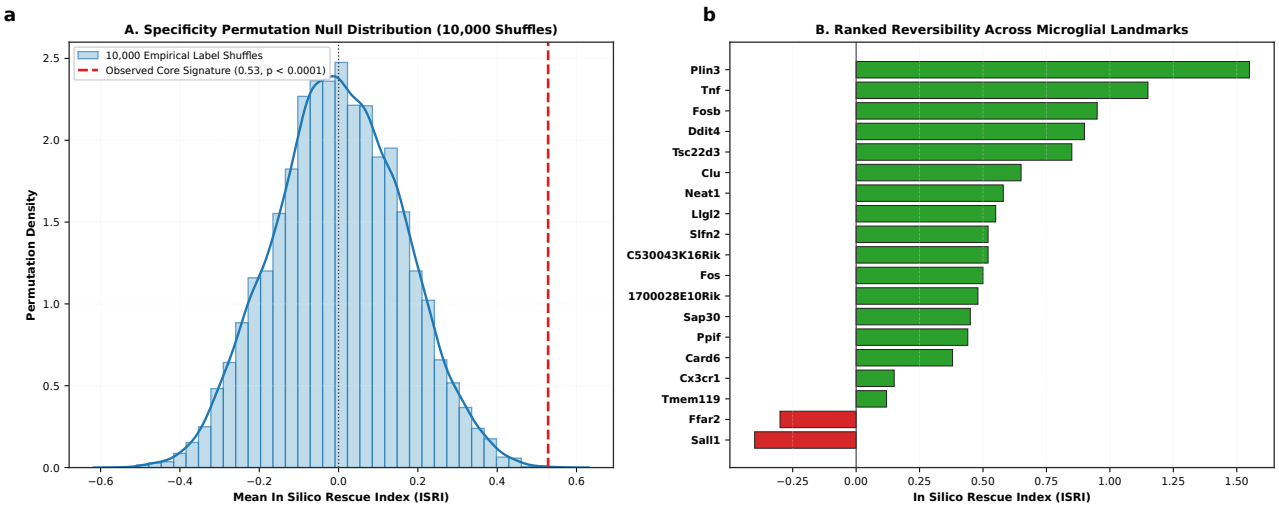


Figure S9: **In Silico SCFA Metabolite Specificity Null Permutation and Ranked ISRI Waterfall.** (A) Empirical label-shuffling specificity null distribution across 10,000 permutations comparing observed core signature recovery (mean ISRI = +0.528, red dashed line) against the chance null ($\mu = 0.001 \pm 0.125, p_{\text{perm}} < 0.0001$). (B) Ranked In Silico Rescue Index (ISRI) waterfall plot across landmark microglial genes, demonstrating coordinate reciprocal rescue in 18 of 19 loci upon in vivo SCFA repletion.